

MRM dataset fetchers and bundled samples

Overview

The MRM empirical callables (`mrm_otis_*`, `mrm_tps_*`, `mrm_siu_*`) operate on three external data sources. This vignette documents how `morie` makes each of them accessible to end users:

- **OTIS** (Ontario Tracking Information System) — public CKAN release on `data.ontario.ca`, reached via the existing `morie_load_dataset()` infrastructure.
- **TPS** (Toronto Police Service) — public ArcGIS Open Data, reached via `morie_fetch_tps()`.
- **SIU** (Ontario Special Investigations Unit) — public Director's Reports, scraped on demand by `morie_fetch_siu()` (the corpus is not shipped because redistribution licensing is unsettled).

Four small reference samples are bundled with the package (`inst/extdata/`) so every example runs offline.

```
library(morie)
#>
#> Attaching package: 'morie'
#> The following objects are masked _by_ '.GlobalEnv':
#>
#> anova_one_way, ask_percy, audit_public_outputs, bootstrap_sample,
#> build_outputs_manifest, build_prompt, buttbp, buttbs, butthp,
#> buttlp, calculate_ebac, calculate_ipw_weights, calibration_weights,
#> canonicalize_cpads_data, chi_square_test, cluster_sample, cohens_d,
#> compare_nested_logistic_models, compute_design_weights,
#> cpads_contract, cramers_v, default_synthetic_name_map,
#> default_workflow_map, design_effect, e_value,
#> effective_sample_size, estimate_aipw, estimate_atc, estimate_ate,
#> estimate_att, estimate_cate, estimate_g_computation, estimate_gate,
#> estimate_irm, estimate_late, estimate_propensity_scores,
#> eta_squared, find_project_root, fisher_exact_test,
#> generate_synthetic_data, hedges_g, hfd, hurst_r,
#> infer_measurement_level, inspect_output, is_over_legal_limit,
#> jackknife_estimate, kendall_tau, kruskal_wallis_test, levene_test,
#> list_morie_modules, load_cpads_data, mann_whitney_test,
#> morie_builtin_db, morie_cache_file, morie_cache_list,
#> morie_cache_load, morie_cache_store, morie_dataset_catalog,
#> morie_dataset_info, morie_db_connect, morie_fetch_ckan,
#> morie_list_datasets, morie_load_cpads, morie_load_dataset,
#> morie_paths, morie_userguide, mrm_classify_mandela, odds_ratio_ci,
#> omega_squared, one_sample_t_test, paired_t_test, pcg_filter,
#> point_biserial_r, power_prop_test, power_t_test, pps_sample,
#> profile_dataset, proportion_ci, read_outputs_manifest,
#> risk_difference_ci, risk_ratio_ci, run_ebac_selection_ipw_analysis,
#> run_morie_module, run_morie_modules, run_pipeline,
#> run_propensity_ipw_analysis, run_treatment_effects_analysis,
```

```
#> run_weighted_logistic_analysis, run_workflow_step,
#> sample_size_logistic, sensitivity_rosenbaum, sgolay_smooth,
#> shapiro_wilk_test, simple_random_sample, spearman_rho,
#> stratified_sample, suggest_analysis_plan, summarize_output_audit,
#> two_sample_t_test, validate_cpads_data, validate_outputs_manifest,
#> verify_statistical_output, wilcoxon_signed_rank_test,
#> write_synthetic_data
```

Bundled reference samples

The samples live in `inst/extdata/` and total ~420 KB. Each is a 1000-row random draw (seed 42) except `otis_b09` and `otis_c11`, which are shipped whole (already small).

```
# Bundled samples
b01 <- morie_sample("otis_b01")
b09 <- morie_sample("otis_b09")
c11 <- morie_sample("otis_c11")
tps <- morie_sample("tps_assault")

# Schema sanity
str(b01, max.level = 1)
#> 'data.frame': 1000 obs. of 18 variables:
#> $ EndFiscalYear : int 2025 2023 2023 2023 2024 2024 2025 2
#> $ UniqueIndividual_ID : chr "2025-08960-SG" "2023-11152-SG" "202
#> $ Gender : chr "Male" "Male" "Female" "Male" ...
#> $ Region_AtTimeOfPlacement : chr "Eastern" "Northern" "Central" "Cent
#> $ Region_MostRecentPlacement : chr "Eastern" "Northern" "Central" "Cent
#> $ Age_Category : chr "25 to 49" "25 to 49" "25 to 49" "25
#> $ NumberConsecutiveDays_Segregation : int 4 1 6 1 2 1 1 1 2 3 ...
#> $ SegReason_SecurityOfInstitution_SafetyOfOthers : chr "No" "No" "No" "No" ...
#> $ SegReason_InmateNeedsProtection : chr "No" "Yes" "No" "No" ...
#> $ SegReason_InmateNeedsProtection_Medical : chr "Yes" "No" "No" "No" ...
#> $ SegReason_SecurityOfInstitution_SafetyOfOthers_Medical : chr "No" "No" "Yes" "Yes" ...
#> $ SegReason_Disciplinary_Segregation : chr "No" "No" "No" "No" ...
#> $ SegReason_InmateRefuseSearch_Scan : chr "No" "No" "No" "No" ...
#> $ MentalHealth_Alert : chr "Yes" "No" "Yes" "No" ...
#> $ SuicideRisk_Alert : chr "Yes" "No" "Yes" "No" ...
#> $ SuicideWatch_Alert : chr "Yes" "No" "Yes" "No" ...
#> $ SegReason_Other : chr "No" "" "" "" ...
#> $ Number_Of_Placements : int 1 8 1 1 1 2 2 1 2 1 ...
nrow(b09); nrow(c11)
#> [1] 78
#> [1] 33
ncol(tps)
#> [1] 31
```

OTIS via CKAN (data.ontario.ca)

The OTIS public release is hosted at data.ontario.ca/dataset/data-on-inmates-in-ontario as 28 CSV resources. The MORIE catalog has four of them registered with their canonical CKAN resource IDs:

```

cat <- morie_dataset_catalog()
cat[cat$source == "otis", c("key", "name", "table_name", "ckan_resource_id")]
#>      key                                     name
#> 37 otisa01      OTIS a01: Restrictive Confinement - Detailed Dataset
#> 38 otisb01      OTIS b01: Segregation - Detailed Dataset
#> 39 otisb09 OTIS b09: Individuals in Segregation - Number of Placements
#> 40 otisc11 OTIS c11: Individuals in Segregation/RC by Aggregate Length
#>      table_name      ckan_resource_id
#> 37  otisa01 5a0c5804-a055-4031-9743-73f556e43bb4
#> 38  otisb01 406e6d90-d568-4553-8ca7-bc9f90e133b9
#> 39  otisb09 df24e943-d52b-43a8-a10e-a3cc906e26bb
#> 40  otisc11 9c7b74a5-53ad-4ef0-a7a6-97772cd01c55

```

To pull the full b01 (82,001 rows) from CKAN:

```

b01_full <- morie_load_dataset("otisb01")
nrow(b01_full)
# 82001

```

The loader downloads the CSV on first call, caches it into the package SQLite database, and returns a data.frame on subsequent calls.

TPS via ArcGIS Open Data

Toronto Police Service publishes per-category crime events through ArcGIS Online. MORIE knows the layer URLs for nine categories.

```

morie_tps_layer_urls()
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Assault_Over_18/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Auto_Theft_Over_18/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Bicycle_Thefts_Over_18/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Break_and_Enter_Over_18/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Homicides_Open_Data_ASR_RC/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Robbery_Over_18/MapServer"
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Shooting_and_Firearm_Discharges_Over_18/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Theft_From_Motor_Vehicle_Over_18/MapServer"
#>
#>      "https://services.arcgis.com/S9th0jAJ7bqgIRjw/arcgis/rest/services/Theft_Over_18/MapServer"

```

To fetch a single category:

```
csv_path <- morie_fetch_tps("Assault")
assault <- utils::read.csv(csv_path)
nrow(assault)
# 254378 (as of mid-2026)
```

The fetcher pages through /query with a 2000-record-per-page cap (the ArcGIS-imposed maximum) and writes a tidy CSV to ~/.cache/morie/tps/tps_Assault.csv. Subsequent calls return the cached path unless `overwrite = TRUE`.

Filtering at the server side:

```
recent <- morie_fetch_tps("Homicides",
  where = "OCC_YEAR >= 2024",
  overwrite = TRUE)
```

SIU via on-demand scraper

The Ontario SIU publishes Director's Reports at siu.on.ca/en/case_directors_reports.php. MORIE includes an on-demand scraper that:

- Iterates over the index page(s) (optionally filtered by year)
- Extracts case links
- Pulls each case detail page
- Parses incident date, notifying police service, Director's decision, and outcome text into a single CSV

```
csv_path <- morie_fetch_siu()      # full unfiltered index
siu <- utils::read.csv(csv_path)
nrow(siu)

# Or restricted to specific years:
csv_path <- morie_fetch_siu(years = 2020:2025, overwrite = TRUE)
```

Why a scraper rather than a shipped dataset?

The legal status of redistributing a single tabular copy of public oversight reports is not clearly established. Running the scraper per-user is unambiguously fair use of public information; bundling the scraped corpus might be more questionable. The scraper itself respects a 2-second rate limit, sets a clear User-Agent, and follows the SIU site's published structure.

R-side wrapper, Python-side implementation

`morie_fetch_siu()` in R is a thin `reticulate` wrapper around `morie.siu_fetch.fetch_siu_cases()` in Python. This keeps the regex parsing logic in one canonical place. If `reticulate` isn't installed, fall back to calling the Python directly:

```
python3 -c "from morie.siu_fetch import fetch_siu_cases; print(fetch_siu_cases())"
```

Workflow: from bundled sample to verified result

Putting it together — a complete worked example without any network call:

```
b01 <- morie_sample("otis_b01")

# Mandela classification with default "individual_any" denominator
mrm_classify_mandela(b01, denominator = "row")
#>   year denominator n_mandela      rate pct n_broader_rc rate_broader
#> 1  2023           362         0 0.000000000 0.00           0 0.000000000
#> 2  2024           337         3 0.008902077 0.89           3 0.008902077
#> 3  2025           301         4 0.013289037 1.33           4 0.013289037
#> 4 pooled          1000         7 0.007000000 0.70           7 0.007000000

# Segregation duration KM
mrm_otis_seg_duration_km(b01,
                          group_cols = "MentalHealth_Alert")
#>   stratum  n mean_days median_days q25_days pct_above_mandela
#> 1    No 499      2.76           2           3           0.6
#> 2    Yes 501      3.29           2           4           0.8
#>   median_among_above_mandela
#> 1                      54.0
#> 2                      41.5

# Mortification co-occurrence: Cramer's V across alert pairs
mrm_otis_mortification_cooccurrence(b01)
#>   alert_a      alert_b    n  chi2 df  p_value crammers_v
#> 1 MentalHealth_Alert SuicideRisk_Alert 1000 33.25 1 8.12e-09 0.1823
#> 2 MentalHealth_Alert SuicideWatch_Alert 1000 12.09 1 5.08e-04 0.1099
#> 3 SuicideRisk_Alert SuicideWatch_Alert 1000 470.37 1 2.66e-104 0.6858
```

For a full-data version, swap `morie_sample("otis_b01")` for `morie_load_dataset("otisb01")` and re-run the same callables.

Caching layout

Cache path	Populated by	Size (full)
~/.cache/morie/morie.db (SQLite)	<code>morie_load_dataset(*)</code>	a few MB to ~1 GB depending on selection
~/.cache/morie/tps/tps_*.csv	<code>morie_fetch_tps()</code>	~5–50 MB per category
~/.cache/morie/siu/SIU.csv	<code>morie_fetch_siu()</code>	~5 MB

References

- OTIS data dictionary — data.ontario.ca/dataset/data-on-inmates-in-ontario
- Toronto Police Open Data — data.torontopolice.on.ca/
- SIU Director's Reports — siu.on.ca/en/case_directors_reports.php
- MRM theoretical paper — Zenodo DOI 10.5281/zenodo.20096075